

Exercise 1.1

1. Find the rank of following matrices.

a. $\begin{pmatrix} -1 & 2 & -2 \\ 4 & -3 & 4 \\ -2 & 4 & -4 \end{pmatrix}$

b. $\begin{pmatrix} 1 & -2 & 3 & 4 \\ -2 & 4 & -1 & -3 \\ -1 & 2 & 7 & 6 \end{pmatrix}$

c. $\begin{pmatrix} 2 & 4 & -4 & 3 \\ 1 & -2 & -1 & 1 \\ 1 & 2 & -1 & 3 \end{pmatrix}$

d. $\begin{pmatrix} 1 & 3 & -2 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{pmatrix}$

e. $\begin{pmatrix} 2 & 1 & 3 \\ 4 & 7 & 13 \\ 4 & -3 & -1 \end{pmatrix}$

f. $\begin{pmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{pmatrix}$

2. Solve by Gauss elimination method:

a. $3x - 2y = 8, 5x + 3y = 7$

b. $2x - y = 5, x - 2y = 1$

c. $x - 2y - 3z = -1, 2x + y + z = 6, x + 3y - 2z = 13$

d. $9y - 5x = 3, x + z = 1, z + 2y = 2$

e. $x + y + z = 6, x - y + z = 2, 2x + y - z = 1$

f. $x - 2y - z = -7, 2x + y = z = 0, 3x - 5y + 8z = 13$

g. $2x + 6y = -11, 6x + 20y - 6z = -3, 6y - 18z = -1$

h. $2x + y + 4z = 1, x + y + z = 1, 4x + y + 10z = 1$

3. Test the consistency and solve.

a. $x + y + z = 6, x - y + z = 8, 5x - 9y = -3$

b. $x + y + z = -2, 2x + 8y + 5z = 5, x + 2y - z = 2$

c. $x + 2y - 3z = 9, 2x - y + 2z = -8, 3x - y - 4z = 3$

d. $x + z = 1, 2y + z = 2, 5x - 9y + 3 = 0$

e. $x - y = 1, z + x = -6, x + y - 2z = 3$

4. Test the consistency and solve.

a. $x + y - 3z = -1, 2x - y + 3z = 4, 15x - 3y + 9z = 21$

b. $x + y - 2z = -2, x - 2y + z = 1, -2x + y + z = 1$

5. Test the consistency and solve if has a solution.

a. $x + 2y - 3z = -1, 3x - y + 2z = 7, 5x + 3y - 4z = 2$

b. $x - 2y + z = 1, x + y - 2z = 1, -2x + y + z = 1$

6. Determine the value of p and q for which the system of linear equations

$x + y + z = 6, x + 2y + 5z = 10, 2x + 3y + pz = q$ has

a. a unique solution

b. no solution

c. infinitely many solutions

Answers

1. a. $\rho(A) = 2$ b. $\rho(A) = 2$ c. $\rho(A) = 3$

d. $\rho(A) = 3$ e. $\rho(A) = 2$ f. $\rho(A) = 3$

2. a. $x = 2, y = -1$ b. $x = 3, y = 1$ c. $x = 2, y = 3, z = -1$

d. $x = 3, y = 2, z = -2$ e. $x = 1, y = 2, z = 3$

f. $x = -2, y = 1, z = 3$

g. No solution h. $x = -3k, y = 2k + 1, z = k$

3. a. $x = \frac{-12}{5}, y = -1, z = \frac{47}{5}$ b. $x = -3, y = 2, z = -1$

c. $x = -1, y = 2, z = -2$

d. $x = 3, y = 2, z = -2$ e. $x = -2, y = -3, z = -4$

4. a. $x = 1, y = 3k - 2, z = k$ b. $x = k - 1, y = k - 1, z = k$

5. a. No solution b. No solution

6. a. $p \neq 6, q \neq 16$ b. $p = 6, q \neq 16$ c. $p = 6, q = 16$

Unit 2

Vector Space

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- addition of vectors and multiplication by scalars
- matrices and their multiplication
- row and column matrices as vectors
- solution of system of linear equations
- rank and system of linear equations

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- different representations of vectors.
- vector spaces.
- $\mathbb{R}^2$ and $\mathbb{R}^3$ as vector spaces.
- vector subspaces.
- linear independent and dependent set of vectors.
- span of vectors.
- basis of a vector subspace.
- linear transformations.
- matrix multiplication over vectors as linear transformation.
- eigen values and eigen vectors of a matrix.
- similarity operations on matrices.
- diagonalization of matrix.
- Cayley Hamilton theorem and its applications.

If following system has non-trivial solution
Find it

$$\textcircled{a} \quad 2x_1 - 5x_2 + 8x_3 = 0$$

$$-2x_1 - 7x_2 + x_3 = 0$$

$$4x_1 + 2x_2 + 7x_3 = 0$$